TASK SPECIFICATION: P8-02 PHASE P8 — SECURITY HARDENING, COMPLIANCE & QUALITY ASSURANCE

# Comprehensive End-to-End Test Suite (Playwright)

Git Branch: `test/P8-02-e2e-suite` · Target Pull Request: test/P8-02-e2e-suite

**Core Objective & Technical Scope:**
Automated browser test suite verifying core user journeys across Arabic and English locales and dual light/dark themes.

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|---|---|---|---|
| Priority Level | High | Complexity Rating | L |
| Execution Status | Backlog 📋 | Target PR / Commit | `test/P8-02-e2e-suite` |
| Upstream Dependencies | P7-06 | | |
| Architecture Blockers | None (Ready for execution) | | |

## IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Configure Playwright test harness with authentication fixtures.
2. Write E2E test specs for parking booking, tutoring enrollment, and tuition payment.
3. Test bilingual switching (AR/EN) and dark/light theme persistence.
4. Integrate Playwright suite into GitHub Actions CI pipeline.

## TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/e2e/**`

## ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- 100% pass rate across all critical user journeys on Chromium, Firefox, and WebKit.
- Automated video and trace capture on failure.

## VERIFICATION & TEST COVERAGE STRATEGY

- Playwright E2E full test suite